

The circumference of a circle

$C = \pi d$ — measured with thread, then trusted as a formula.

LESSON 92–94

3 × 40 MIN

MATEMATIKA 6, §19

S7 8.2

LESSON CLOCK

25'

30'

35'

25'

5

The measurement	25 min
The formula	30 min
Backwards	35 min
Wheels and edges	25 min
Homework	5 min

LEARNING OBJECTIVES

- Find the ratio $C \div d$ by measurement.
- Use $C = \pi d$ and $C = 2\pi r$.
- Find the radius or diameter from the circumference.
- Solve practical problems about wheels and circular edges.

TEXTBOOK

National	pp. 262–272
Cambridge	Stage 7, pp. 80–84
Workbook	Exercise 8.3

01

Explanation

The measurement

Wrap thread round several round objects and divide the length by the diameter.

OBJECT	D	C	$C \div D$
a coin	2.4 cm	7.5 cm	3.13
a glass	7 cm	22 cm	3.14
a plate	21 cm	66 cm	3.14
a bucket	28 cm	88 cm	3.14

THE SAME NUMBER, EVERY TIME

Whatever the size of the circle, the ratio lands near 3.14. That constant is π , and the small differences in the table are measuring error, not differences between the circles.

The formula

$$C = \pi d = 2\pi r$$

R	D	C WITH $\pi = 3.14$	WITH $\pi = \frac{22}{7}$
5	10	31.4	31.43
7	14	43.96	44
10	20	62.8	62.86
21	42	131.88	132

CHOOSE THE APPROXIMATION TO SUIT THE NUMBERS

$\frac{22}{7}$ is exact-looking whenever the radius is a multiple of 7; 3.14 suits everything else. Both are approximations to the same constant.

Backwards

GIVEN	WORKING	ANSWER
$C = 31.4$	$d = C \div \pi$	$d = 10$
$C = 44$	$r = C \div 2\pi$	$r = 7$
$C = 62.8$	$r = C \div 2\pi$	$r = 10$
$C = 132$	$d = C \div \pi$	$d = 42$

DIVIDING BY π GIVES THE DIAMETER, NOT THE RADIUS

$C \div \pi = d$, and the radius is half of that. Stopping one step early is the standard error in the reverse question.

Wheels and edges

PROBLEM	WORKING	ANSWER
a wheel of diameter 70 cm: one turn	$\frac{22}{7} \cdot 70$	220 cm
how many turns in 220 m	$22\ 000 \div 220$	100
a round pond of radius 14 m: the fence	$2 \cdot \frac{22}{7} \cdot 14$	88 m
a running track of radius 35 m: one lap	$2 \cdot \frac{22}{7} \cdot 35$	220 m
a bicycle wheel of radius 35 cm in 50 turns	$50 \cdot 220$	110 m

KEEP THE UNITS IN ONE SYSTEM

Turning 220 m into 22 000 cm before dividing avoids the factor-of-a-hundred error that this kind of problem invites.

02

Worked examples

EXAMPLE 1 A circle has radius 7 cm. Find its circumference, taking $\pi = \frac{22}{7}$.

1

$C = 2\pi r$

2

$= 2 \cdot \frac{22}{7} \cdot 7$

3

$= 44\text{ cm}$

ANSWER

44 cm

EXAMPLE 2 A circle has circumference 62.8 cm . Find its radius, taking $\pi = 3.14$.

① $C = 2\pi r$

② $62.8 = 6.28r$

③ $r = 10\text{ cm}$

ANSWER

10 cm

EXAMPLE 3 A wheel of diameter 70 cm rolls 220 m . How many turns?

① One turn: $\frac{22}{7} \cdot 70 = 220\text{ cm}$.

② 220 m is $22\,000\text{ cm}$.
Same units.

③ $22\,000 \div 220 = 100$

ANSWER

100 turns

03 Interactive model

Do the thread measurement in groups and tabulate every ratio on the board; π arrives as a discovery rather than a fact to be believed.

Circumference, forwards and backwards

$C \div d$ equals:

With $r = 7$ and $\pi = \underline{22} \ 7$, C is:

With $d = 10$ and $\pi = 3.14$, C is:

$C = 62.8$ gives r equal to:

5

10

20

31.4

$C = 44$ gives r equal to:

7

14

22

3.5

A wheel of diameter 70 cm covers per turn:

70 cm

140 cm

220 cm

440 cm

In 220 m it makes:

10 turns

100 turns

1000 turns

22 turns

$C \div \pi$ gives:

- the radius
- the diameter
- the area
- π

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	ЎЗБЕКЧА	РУССКИЙ
Circumference	Aylana uzunligi	Длина окружности
Pi	Pi soni	Число пи
Diameter	Diametr	Диаметр
Radius	Radius	Радиус
Ratio	Nisbat	Отношение
Revolution	Aylanish	Оборот
Approximation	Taqribiy qiymat	Приближение
Formula	Formula	Формула

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

π is the ratio of:

C to r

C to d

d to r

r to C

C equals:

πr

πd

πr^2

$2\pi d$

A circle of radius 5 has circumference about:

15.7

31.4

78.5

10

To find d from C you:

multiply by π

divide by π

divide by 2π

square it

A pond of radius 14 m needs a fence of about:

44 m

88 m

176 m

616 m

22 7 is best used when the radius is a multiple of:

2

3

7

10

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $r = 5$: the circumference ($\pi = 3.14$)

ANSWER 31.4

2. $r = 7$: the circumference ($\pi = \frac{22}{7}$)

ANSWER 44

3. $d = 20$: the circumference ($\pi = 3.14$)

ANSWER 62.8

4. $d = 42$: the circumference ($\pi = \frac{22}{7}$)

ANSWER 132

5. $C \div d$

ANSWER π

6. $r = 10$: the circumference

ANSWER 62.8

7. $r = 21$: the circumference ($\pi = \frac{22}{7}$)

ANSWER 132

Medium · the standard question

7 PROBLEMS

1. $C = 31.4$: the diameter

ANSWER 10

2. $C = 44$: the radius

ANSWER 7

3. $C = 62.8$: the radius

ANSWER 10

4. $C = 132$: the diameter

ANSWER 42

5. A wheel of diameter 70 cm: one turn

ANSWER 220 cm

6. Turns in 220 m

ANSWER 100

7. A pond of radius 14 m: the fence

ANSWER 88 m

Hard · stretch and reason

7 PROBLEMS

1. A track of radius 35 m: one lap

ANSWER 220 m

2. Laps in 1 100 m

ANSWER 5

3. A bicycle wheel of radius 35 cm in 50 turns

ANSWER 110 m

4. A circle whose circumference equals its diameter times 3.14 has radius 9: its C

ANSWER 56.52

5. Doubling the radius does what to the circumference?

ANSWER Doubles it

6. A circular table of circumference 220 cm: its diameter

ANSWER 70 cm

7. Why is $C \div d$ the same for every circle?

ANSWER All circles are the same shape, scaled

07 Homework

Homework — 5 tasks

Say which value of π you are using, and why.

1. Find the circumference of circles with $r = 3.5$ cm and $d = 28$ cm.
2. Find the radius of a circle with circumference 88 cm.
3. A wheel of diameter 56 cm makes 100 turns. How far does it go?
4. A circular flower bed has radius 7 m. Find the length of edging needed.
5. Measure three round objects at home and tabulate $C \div d$ for each.